Multitemporal Hyperspectral Images Change Detection Based on Joint Unmixing and Information Coguidance Strategy

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***Abstract*— The richness of spectral information in multitemporal hyperspectral images (MHSIs) offers the possibility to effectively detect subtle changes and properties of grounds. However, severe spectral variabilities and inadequate spatial co-exploitation capabilities limit the performance of existing methods due to differences in acquisition times and conditions. Therefore, this article proposes a strategy of joint unmixing and multitemporal spatial information coguidance (JUC) to fully exploit the spatio-temporal-spectral features. First, a multitemporal joint unmixing method is used to achieve endmembers' extraction and abundance estimation. Wherein the method adds spectral perturbed regularization when compared to the traditional unmixing strategy, making it robust to spectral variability. Second, we propose a multitemporal coguidance method that highlights the contrast between changed and unchanged regions and removes the noise by transferring the common structure information between the multitemporal first principal component map and the abundance difference maps. It will obtain an enhanced abundance difference maps and achieve effective combination of multitemporal spatial information. The final change result can be obtained by combining and thresholding these enhanced abundance difference maps. Experiments on some data sets demonstrate that the proposed algorithm is sufficiently valid and robust for multitemporal images, especially for data containing spectral variabilities and obvious structures.**

***Index Terms*— Change detection (CD), hyperspectral image, joint unmixing, multitemporal information coguidance.**

I. Introduction

IN ADDITION to its rich spectral information, multitemporal hyperspectral images (MHSIs) also have unique temporal information that enables them to detect subtle changes and identify the changing properties. This is complicated by the high data dimensionality, the widespread presence of mixed pixels, and the inefficiency of spatio-temporal information association. The high dimensionality of MHSIs increases the computing complexity and time; the presence of mixed pixels will also bring some commission and omission errors [1]. These errors will severely affect the detection accuracy; and the methods of spatio-temporal information utilization are relatively limited as well. All of these issues may lead to some methods being poorly applied.

Currently, the main methods of change detection (CD) include both pixel-level and subpixel-level methods. References [2] and [3] are pixel-level methods, which are transferred from multispectral image analysis and focus on analyzing multitemporal images from a statistical learning perspective. An enhanced principal component analysis and change vector analysis (PCA–CVA) method [4] is utilized to identify the changes between bitemporal hyperspectral images based on inverse triangular function. However, it is implemented based on the assumption that pure-pixel theory and the spectral utilization is relatively low. It is well known that the mixed pixels are a common phenomenon in most cases due to the limitation of spatial resolution. Hence, pixel-level methods like above may lose subtle changes in spectrum. To fully exploit the characteristics of hyperspectral image, the idea of subpixel-level processing is applied to CD.

The subpixel-level methods are mainly based on the idea of spectral unmixing to detect subtle changes while obtaining significant changes. Spectral unmixing is to determine the reference spectral features composing a hyperspectral image and the corresponding abundance fractions in each pixel [5]. Based on this, many methods have been proposed in recent years. The method [1] has been presented to detect the multiple types of changes based on a block-based unmixing strategy, which first stacks the images and divides into blocks; then, the endmembers of each block can be extracted; further, the unmixing process is carried out based on this endmember set. This processing way will further improve the effectiveness of the endmember extraction and increase the detection accuracy. Meanwhile, a sequential spectral CVA (SSCVA) that considers spectral continuity [6] and a hierarchical method [7] are utilized to identify the properties of changes. They all explore the spectral features in depth to analyze MHSIs. However, the spectral variability between MHSIs of the same endmembers may affect the change analysis. In [8], a sparse-based method is presented to obtain subpixel-level changes. Similar strategies have been proposed in [9] and [10] to address the existing problems of endmembers extraction and dictionary learning. However, they represent endmembers as

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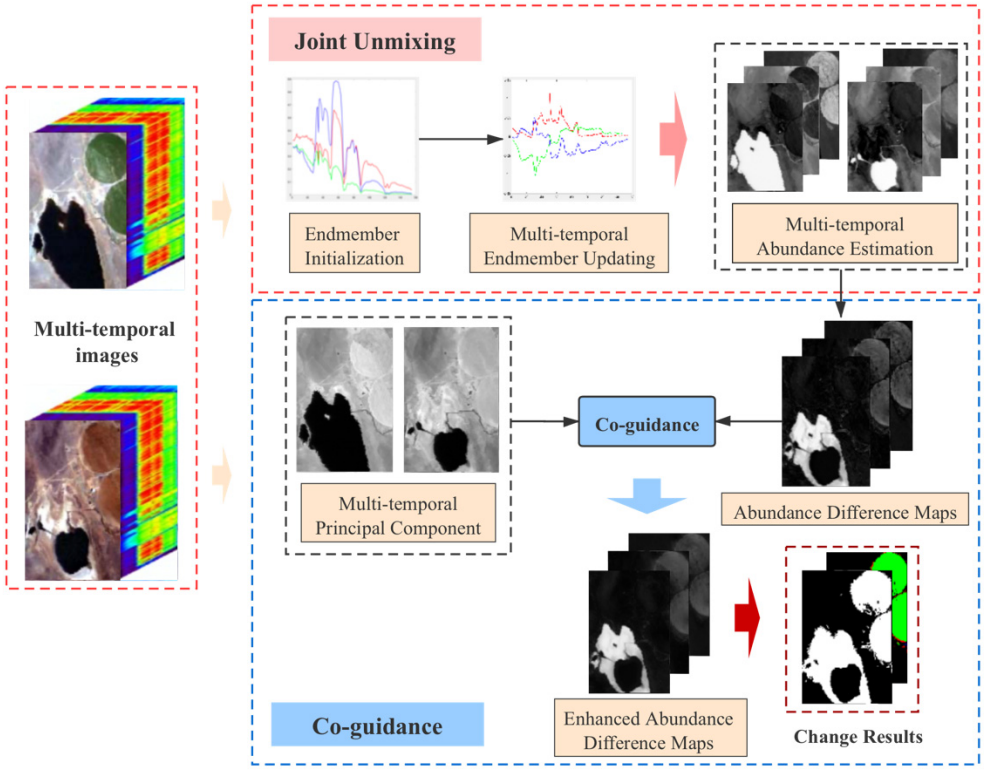


Fig. 1. Framework of the proposed method.

members of spectral libraries associated with each material. In other word, the spectral libraries are needed to be as support. Therefore, the absence of a particular object spectrum may limit the applicability of these methods.

Deep learning-based methods have been exploited to CD as well, because they are said to be better at handling the problem of high dimension and extracting effective features when compared to other traditional methods. A method combining deep belief network and clustering [11] is adopted to extract changes. A novel general end-to-end 2-D convolutional neural networks (CNNs) strategy [12] is presented to combine linear and nonlinear unmixing results to design a new network structure. Additionally, other networks, such as improved CNNs [13], Siamese Convolutional Network [14], and Fully Convolutional Siamese Networks [15], are seen as new feature extractors or similarity metrics to complete CD tasks. However, there are still some issues to be addressed. On the one hand, the problems of the high complexity and computing time are inevitable. While these methods increase intelligence to some extent, the resulting storage costs are also significant. On the other hand, these networks are limited to some application conditions. The premise of network training is that it should adopt a supervised or semisupervised way, whereas it is difficult to obtain enough label samples for the training process of the corresponding method, meaning that unsupervised processes may even more desirable in real applications. Therefore, some unsupervised methods [16]–[18] are explored. They obtain an initial label by automatic extraction method and then select training samples based on this label to train the network. It is to be noticed that these unsupervised ways do not refer to an untrained process, but rather its training samples are not from manual labeling but from some basic detection methods. Network training can be defective when the initial detection method obtains an inaccurate label or has a large bias due to multiple factors, resulting in an accumulation of errors and reduced detection performance.

Thus, it is necessary to design an effective unsupervised CD strategy that not only has a strong ability to utilize multitemporal information, but also can quickly and accurately determine the changing result. In this context, this article proposes a new CD framework for MHSIs, which maintains the advantage of spectral information for the detection of subtle changes, meanwhile, combines the inherent structural knowledge of multitemporal images to achieve a transfer of spatial information. This operation way can not only be robust to spectral variability, but also improve the detection accuracy by making full use of multitemporal spatial information. The main contributions of this article are as follows.

1) We adopt a new joint unmixing method [19], [20], which is robust to spectral variability and transfer it to the realm of CD to enable the extraction of temporal and spectral information.
2) We use a multitemporal spatial information coguidance strategy that injects the common structure between multitemporal first principal component maps and the abundance difference maps to enhance the details and remove the noise, where the guidance process is a completely linear optimization problem that greatly reduces the computational time.

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3) We design a new CD framework that iteratively transfers spatial information from multitemporal images, different from previous ones based on difference hyperspectral or stacked hyperspectral images.

The remainder of this article is organized as follows. Section II describes the proposed framework in detail. Section III reports experimental results using some data sets. The detail discussions are also presented in this section. Finally, conclusions are given in Section IV.

II. Proposed Framework

In this section, we illustrate the proposed CD method thoroughly. For ease of understanding, Fig. 1 displays the flowchart of the proposed method (JUC). It contains two main parts: joint unmixing and coguidance. In the former process, we introduce a joint unmixing strategy that takes spectral variability into account, which makes full use of the spectral information of registered multitemporal images to iteratively update and optimize the endmembers and abundance. Moreover, the abundance difference maps can be obtained by subtracting the corresponding abundance maps of different temporal images. In the coguidance stage, the common structure between the first principal component map of each temporal image and the abundance difference maps of each endmember will be identified to implement guidance enhancement for the abundance difference maps obtained in the first step. The accomplishment of this step will highlight the contrast of between changes and unchanged regions, and remove the noise. The final change results will be obtained by thresholding. The initial endmembers used for iterative updating during the joint unmixing step can be obtained by vertex component analysis (VCA) [21].

Subpixel-level based analysis is one of the important processing ideas for MHSIs CD because of its high spectral utilization and ability to detect many types of subtle changes. However, some existing methods, whether separately unmixing or stacked unmixing, may be affected by spectral variability. These spectral variabilities do not refer to local variations on a single temporal image, but rather to the spectral variability and perturbation that is unavoidable between multitemporal images due to different acquisition time. Multitemporal joint unmixing strategy is to overcome the impact of spectral variabilities and improve the detection accuracy. Additionally, the above unmixing method utilizes only temporal and spectral information, with little consideration of multitemporal spatial information. That is the biggest problem with the current subpixel-level-based methods. The multitemporal spatial information does not refer to pixel neighborhood information on a single image, but rather to the spatial structure between multitemporal images. Combining with these structures and the results of joint unmixing will further improve the detection results.

Next, we will introduce how to achieve the joint unmixing, and what structure of multitemporal images should be considered and how these structures can be used to achieve guidance enhancement.

A. Multitemporal Images Joint Unmixing

The traditional separate linear unmixing models treat a pixel as a summation of the multiplication of endmembers and abundances and do not take into account the effects of spectral variability and perturbations in the whole process. Therefore, these unmixing methods tend to perform poorly in multitemporal images. The multitemporal joint unmixing provides a solution to the above problem. The main steps of joint unmixing can be divided into two parts; first, pick an arbitrary temporal image in the multitemporal sequence and extract the endmembers by VCA; then, select another temporal image from the multitemporal sequence and make full use of the spectral information of this temporal image to update and optimize the endmembers of the previous temporal image so that it can be suitable for the current temporal image. Repeat the second part until all temporal image have been realized for spectral unmixing. The abundance maps of multitemporal images can be obtained. This processing way can achieve the joint analysis of multitemporal images. In this process, only one temporal image and the endmembers of the previous temporal image need to be loaded into memory each time, which greatly reduces the memory consumption. In addition, a spectral perturbation term can be added into the unmixing process to obtain a more robust result.

Assuming that the number of endmembers extracted from the initial temporal image is K, the number of images contained in multitemporal sequence is T, and the number of pixels of each temporal image is N. The unmixing model can be represented as

$$x_{\mathrm{it}} = \sum_{k=1}^{K} a_{\mathrm{kit}}(e_{\mathrm{kt}} + de_{\mathrm{kt}}) + b_{\mathrm{it}} \tag{1}$$

where x_{it} represents the ith pixel in the tth image and $i = 1, \ldots, N$, $t = 1, \ldots, T, k = 1, \ldots, K$. a_{kit} is the abundance value of the kth endmember in the ith pixel of the tth image. e_{kt} denotes the kth endmember of tth image. de_{kt} is the spectral perturbation caused by spectral variability of tth temporal image kth endmember. b_{it} represents the noise of the ith pixel in the tth image, due to data acquisition or model bias. The matrix form is

$$X_t = (E_t + dE_t)A_t + B_t \tag{2}$$

where $X_t = [x_{1t}, \ldots, x_{Nt}]$ is a matrix that contains all the pixels of the tth temporal image. E_t is endmembers matrix where each column represents an endmember. A_t is the abundance matrix. dE_t represents the spectral perturbation matrix where each column represents a perturbation vector. B_t denotes the noise matrix. In the unmixing process, both abundance and endmembers have their own numerical limits that are mentioned in many unmixing methods [1]. Therefore, a similar idea can be used to limit spectral perturbations. The form of limitation is

$$\|dE_t\|_F^2 \le \sigma^2, \quad \text{where } t = 1, \ldots, T \tag{3}$$

$$\left\| \sum_{t=1}^{T} dE_t \right\|_F^2 \le \kappa^2 \tag{4}$$

where σ and κ are constant parameters that control the energy value and instantaneous energy variation of the spectral

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variability, respectively. The values of these two parameters are related to the spectral variability between multitemporal images. Thus, the above unmixing problem is transformed into a problem of solving endmembers M, abundances $\wp_A$, and perturbation parameters P_t, which can be regarded as an empirically minimized stochastic optimization problem [19]. The objective function to be optimized is

$$\min_{E_t \in M} \frac{1}{T} \sum_{t=1}^{T} H(X_t, E_t) + \tau \Psi(E_t) \tag{5}$$

$$H(X_t, E_t) = \min_{(A_t, dE_t) \in \wp_A \times \mathrm{P}_t} F(X_t, E_t, A_t, dE_t) \tag{6}$$

where Ψ represents the regularization term and τ is parameter. F is the normalized instantaneous difference measure function and H denotes the loss when using the tth temporal image to update the endmembers. The above optimization problems can be solved using proximal alternating linearized minimization (PLAM) method. The detailed solution procedure, the impact of individual parameters on the results, and the convergence and computational complexity of the algorithm can be found in [19] and [20]. In the experimental parts, all parameters are selected based on the optimal rules.

B. Multitemporal Spatial Information Coguidance

According to the strategy of endmembers optimization and updating in joint unmixing, the corresponding endmembers set and abundance images can be obtained for each temporal image. Meanwhile, abundance difference maps can be obtained by subtracting abundance image corresponding to the same endmember of different temporal images. Some existing unmixing-based CD methods obtain detection results by thresholding these abundance difference maps. Those approaches generate some false changes because the spatial structure information between the multitemporal images is hardly taken into account in joint unmixing, while the lack of distinct boundary between the changed and unchanged regions also makes the detection result is sensitive to threshold selection. It is worth noting that the mentioned spatial structures are not spatial information of a single image, but rather spatial knowledge between multitemporal images. Hence, the aim in this section is to enhance the abundance difference maps using the multitemporal spatial information and thus address the above existing problems.

To obtain the enhanced abundance difference maps, it is necessary to obtain the image data containing multitemporal spatial structures, determine which structures should be utilized, and complete the coguidance. PCA as a linear transformation can achieve the concentration of energy and its first principal component map can preserve the main spatial structure information of each temporal image. Therefore, it can be considered as image data containing spatial structure [22]. Guided filter is used in the field of classification as a relatively new method of image information-guided injection [23]. It optimizes and updates the information of the target map using information from the reference map to achieve guidance. This idea can be transferred and implemented for the purpose of enhancing the abundance difference maps using the first principal component map. However, during the target map optimization process, the guided filter is fully trusted with the reference map. This means that all the structure in the reference map will be injected into the target map. It is unacceptable in our task, because there are many inconsistent structures contained between the first principal component maps and the abundance difference maps. If these inconsistencies are injected completely into the abundance difference maps, they will not only fail to achieve the purpose of enhancement, but will introduce other false changes. Therefore, the common structure should be searched and injected into the abundance difference maps as the trusted reference information to achieve guidance enhancement.

The common structure search method can quantitatively determine whether the structure is the same in two images. The most commonly used search methods are those based on image blocks, gradient differences within blocks, and the normalized cross correlation coefficient (NCC). The NCC is a quantitative index of structural similarity for which the value is [0, 1], the larger the value, the more structural similarity. However, since it is a nonlinear process, to facilitate optimization, we consider using the relationship between the minimum error in the optimized least-squares regression function and the NCC to transform this search strategy into a linear form. In the following, the first principal component map of one temporal image is used as a reference map and the abundance difference map corresponding to one endmember is used as a target map to model the search and optimization for the common structure. According to the relationship mentioned above, the least-squares regression function needs to be obtained and can be expressed as

$$R\left(I, G, a_p^1, a_p^0\right) = \sum_{q \in N(p)} \left(a_p^1 I_q + a_p^0 - G_q\right)^2 \tag{7}$$

where I is the abundance difference map, G is the first principal component map, and $a = \{a_p^1, a_p^0\}$ represents the weight and bias coefficients. $N(p)$ denotes the block centered at the point $p = (x, y)$. The number of blocks is r. q denotes the index subscript of the pixel within $N(p)$. The minimum error in the optimized least-squares regression function can be represented as

$$e\left(I_p, G_p\right)^2 = \min_{a_p^1, a_p^0} \frac{1}{|N|} R\left(I, G, a_p^1, a_p^0\right) \tag{8}$$

where $|N|$ is total number of pixels in block $N(p)$. $e\left(I_p, G_p\right)^2$ is the error function that measures the similarity within the blocks. It can be found that this function is a linear optimization. We can sum all the blocks and optimize the coefficient a to obtain the common structure. Therefore, the function to be optimized can be expressed as

$$g(I, G, a) = \sum_{p} R\left(I, G, a_p^1, a_p^0\right). \tag{9}$$

By optimizing the above function, the optimal value of the coefficients with minimum error can be obtained to achieve the search of the common structure. However, during the optimization process, an invalid solution will be generated, and when the structure obtained from the invalid solution is

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used to guide the abundance difference map, the enhanced results may deviate from the original abundance difference map significantly. Therefore, to discard this invalid solution in the calculation, the constraint term can be added and be denoted as

$$C(I) = \sum_p \beta \left\| I_p - I_{0,p} \right\| \tag{10}$$

where I_p and $I_{0,p}$ are the pixel value of position p in the enhanced abundance difference maps and the original abundance difference maps, respectively. β represents the constraint parameters. Updating the abundance difference maps by considering the structure search and structure constraints methods will achieve common structure injection. In addition, because the effect of noise will reduce the detection accuracy, guidance methods like guided filter [24], trilateral filter [25], we can introduce a smoothing term to limit these noises, which can be expressed as

$$S(a) = \sum_p \varepsilon_1 \left(a_p^1\right)^2 \tag{11}$$

where ε_1 denotes the smoothness parameter. The value of ε_1 should be slightly smaller to avoid removing the real changes. Therefore, the final objective function to be optimized can be expressed as

$$f(I, G, a) = g(I, G, a) + C(I) + S(a). \tag{12}$$

The enhanced abundance difference maps can be obtained by optimizing *f*. Repeat the above optimization process until all temporal images are guided into the abundance difference maps corresponding to the first endmember. The final enhanced abundance difference map of the first endmember can be obtained by fusing these abundance difference maps, which can be represented as

$$I_{s,k} = \frac{1}{T} \sum_{i=1}^{T} I_{[i,k]} \tag{13}$$

where $I_{[i,k]}$ denotes the enhanced abundance difference map after the ith temporal first principal component map guidance injection corresponding to the kth endmember. $I_{s,k}$ represents the enhanced abundance difference map corresponding to the kth endmember. In these enhanced images, the contrast between changes and unchanged regions will be improved, the structure of changes will be highlighted, and the noise will be removed.

C. Modeling Solutions

The parameter to be solved in the model is a. Similar to principle of guided filter, the coefficient a can iterative optimization and updating of the abundance difference maps. The updated rule of abundance difference maps and the solution of a can be represented as follows, respectively [24]:

$$I_p^{(n+1)} = \frac{1}{M_I^{(n)}} \left(J_n^{(n)} G_p + K_I^{(n)} + \beta I_{0,p} \right) \tag{14}$$

$$a_p^{1(n)} = \frac{\operatorname{cov}\left(I_p^{(n)}, G_p\right)}{\sigma\left(I_p^{(n)}\right) + \varepsilon_1}, a_p^{0(n)} = \mu\left(G_p\right) - a_p^{1(n)} \mu\left(I_p^{(n)}\right) \tag{15}$$

where n represents the number of iterations and $\mu(\cdot)$ denotes the pixel mean value in the block $N(p)$. Meanwhile

$$M_I^{(n)} = \frac{\beta}{|N|} + \mu\left(a_p^{1(n)} \cdot a_p^{1(n)}\right) \tag{16}$$

$$J_I^{(n)} = \mu\left(b_p^{1(n)}\right) + \mu\left(a_p^{1(n)}\right) \tag{17}$$

$$K_I^{(n)} = \mu\left(b_p^{0(n)}\right) - \mu\left(a_p^{1(n)} \cdot a_p^{0(n)}\right) \tag{18}$$

where

$$b_p^{1(n)} = \frac{\operatorname{cov}\left(G_p, I_p^{(n)}\right)}{\sigma\left(G_p\right) + \varepsilon_1}, a_p^{0(n)} = \mu\left(I_p^{(n)}\right) - b_p^{1(n)} \mu\left(G_p\right). \tag{19}$$

We use the solution of the related parameters to replace the detailed solution procedure. The derivative of function f yields the solution of the coefficient a. The update of I can be obtained using the correlation principle of the guided filter and the Jacobi method [26]–[28]. As n increases, the inconsistent structure and noise will be removed, and the final changes will be highlighted. In this process, the common structure is iteratively injected into the abundance difference map, which achieves the purpose of the multitemporal spatial information injection, meanwhile, compensates for the inadequacy of some methods in the spatial information co-exploitation.

D. Abundance Combination and Changes Analysis

The enhanced abundance difference maps can be obtained by joint unmixing and full utilization of multitemporal spatial information. In these maps, the boundary between the changes and unchanged regions is clear, the noise is removed, and the structures of the changes are integrated. Actually, CD can be divided into binary CD and multiclass CD. The former is to identify the positions of changes and unchanges, while the latter is to obtain change classes based on changed positions.

The binary CD result can be obtained by thresholding the enhanced abundance difference maps. It can be expressed as

$$B_{\mathrm{CM}} = \sum_{k=1}^{K} \operatorname{th}\left(I_{[s,k]}\right) \tag{20}$$

where $\operatorname{th}(\cdot)$ represents thresholding and B_{CM} denotes the final binary change map. Thresholds are important parameters used to obtain binary changes in the thresholding process. Generally, the threshold is selected based on the distribution of pixel values in difference images.

The multiclass changes can be obtained by joint unmixing. In fact, in the joint unmixing process, the endmembers satisfy linear independence and the spectrum of different endmember corresponds to that of different change classes [1]. This means that the number of endmembers is the same as the number of change classes. Additionally, according to the correspondence between abundance and endmembers in the linear unmixing process, the number of abundance features and enhanced abundance difference maps is equal and the same as the number of change classes as well. Therefore, multiclass changes can be detected by combining these enhanced abundance difference features. According to the above analysis, the combination way can be summarized. For changes, assuming that each change pixel has the same label on the same enhanced abundance

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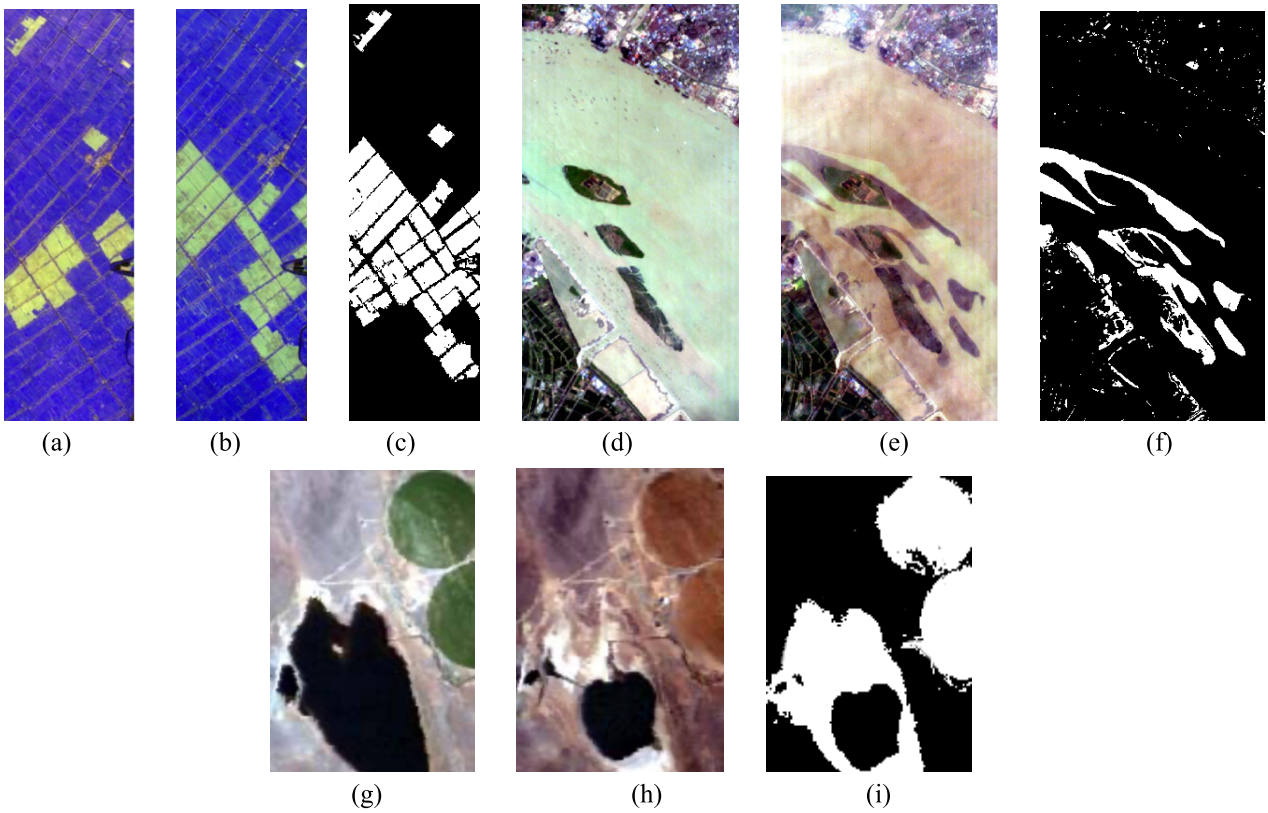


Fig. 2. Co-registered real MHSIs data sets. (a) T1 false color image of the farmland data set acquired on May 3, 2006. (b) T2 false color image of the farmland data set acquired on April 23, 2007. (c) Ground truth of farmland data set. (d) River data set T1 acquired on May 3, 2013. (e) River data set T2 acquired on December 31, 2013. (f) Ground truth of river data set. (g) Lake data set T1 acquired on April 10, 2014. (h) Lake data set T2 acquired on November 17, 2014. (i) Ground truth of lake data set.

difference map, and has different labels on different maps. We can determine the final label of each change pixel by comparing the pixel value at the same coordinates on all enhanced abundance difference maps. The determination criterion is that the label of each change pixel is the same as the label of the enhanced abundance difference map where the largest pixel value is located, which can be expressed as

$$M_{\mathrm{CM}}(i, j) \in \arg\max_{c_k \in C} \left(I_{s,k}(i, j) \right) \tag{21}$$

where the number of change types is K, k is the index, and $k = 1, \ldots, K$. $C = \{c_1, \ldots, c_K\}$ represents the set of change types, c_k is the kth class. i and j are index of the enhanced abundance difference maps $I_{s,k}$, which have different labels.

III. Experimental Results and Discussion

To evaluate the performance of the proposed method (JUC), several experiments are conducted on some real data sets that are preprocessed by radiometric and registration. The first farmland data set was taken by Earth Observing-1 (EO-1) Hyperion from Jiangsu, China, with a size of 450 × 140 pixels and 155 bands, which were obtained on May 3, 2006, and April 23, 2007, respectively. The second river data set was acquired on May 3, 2013, and December 31, 2013, respectively, by EO-1 Hyperion from Jiangsu, China, with a size of 463 × 241 pixels and 198 bands. The main changes appear in a river. It contains many complicated situations, such as illumination and seasonal differences [12]. The third multitemporal data set was obtained by Airborne Visible Infrared Imaging Spectrometer (AVIRIS) over the Lake Tahoe (California, USA) between 2014. The size of this data set is 150 × 110 pixels and 173 bands. It also contains complicated changes because of serious illumination and seasonal differences. The color images and the corresponding ground truths marked by visual interpretation are displayed in Fig. 2.

A several outstanding unsupervised methods have been used to compare with JUC, including pixel-level method, such as PCA–CVA [4] and many subpixel-level methods, such as sparse unmixing (SU) [8], binary and multiclass analysis (BM) [30], and joint unmixing without coguidance (JUWC). PCA–CVA is an important pixel-level method that detects changes based on optimizing inverse triangular function. According to our previous analysis, PCA–CVA is able to achieve the detection of distinct changes, but may not be able to detect subtle changes in some cases, as its utilization of spectral information needs to be improved. The purpose of using this method is to further validate that subpixel-level methods may be more effective for MHSIs CD. SU is an interesting subpixel-level method to detect the subtle changes based on spectral libraries [8], which use Variable Splitting Augmented Lagrangian and Total Variation (SUnSAL-TV) method to acquire endmembers and abundance images, respectively, while later thresholding and combining the corresponding abundance difference maps. BM is a subpixel-level method to

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TABLE I
QUANTITATIVE COMPARISON OF FARMLAND DATA SET CD RESULTS BY DIFFERENT METHODS

Method	Precision	Recall	F-score
PCA-CVA	96.10%	85.86%	90.69%
SU	95.75%	94.35%	95.04%
BM	87.15%	80.16%	83.51%
JUWC	95.71%	93.23%	94.45%
The proposed	**96.86%**	**94.97%**	**95.39%**

achieve CD as well, according to three constraint conditions to obtain binary changes and multiclass changes. JUWC is our improved method within the existing framework, which uses only joint unmixing taking into account spectral variability to obtain abundance difference maps and then thresholding and combining these difference maps. In this method, the joint unmixing strategy corresponds to the Section II-A. The introduction of this method is to highlight the effectiveness of the proposed coguidance strategy in addition to verifying the conclusion that the joint unmixing method is robust to spectral variability. The performance of those methods is evaluated by precision, recall and F-score [31]. Higher precision and recall values indicate small number of false alarms and missed detections, respectively. Meanwhile, F-score is evenly balanced between false alarms and missed detections, revealing the overall detection performance [32]. All experiments are implemented on a desktop computer with 2.90-GHz CPU and 32 GB of RAM.

A. Results and Discussion of Farmland Data Set

Since Farmland data set is captured in the same season and at the same angle of view, the spectral variability across each temporal image may have little effect. It mainly consists of changes in the types of crops and growth states. The whole experiment can be divided into two parts: joint unmixing and coguidance. In the parameter setting for joint unmixing, the number of endmembers is four that is estimated by empirical automatic estimation (EAE) method [33] and all other parameters will be set according to the optimal rules [19]; in the later step, the parameter is set to: $r = 3$, $\varepsilon_1 = 10^{-4}$, $\beta = 30$, $n = 30$, and th $= 0.15$. The parameters of the other comparison methods are set using the optimal parameters given in this article. The running time of PCA-CVA, SU, BM, JUWC, and JUC is 5, 60, 35, 10, and 20 s, respectively. The specific experimental results are shown in Figs. 3 and 4 and Table I.

The respective advantages of joint unmixing and coguidance are shown in Fig. 3. Comparing the abundance results [Fig. 3(a)–(d)] of the first and fourth endmembers, it can be found that the spectral difference between the corresponding abundances is relatively small under the joint unmixing condition. Therefore, the abundance difference maps may contain less false changes brought by spectral variability [Fig. 3(e) and (g)]. This is the advantage bring about by the joint unmixing. In addition, the storage consumption is relatively low since the adopted joint unmixing scheme is a process of dynamically optimizing the endmembers set. By comparative observation of Fig. 3(e)–(h), it can be found that the coguidance injection of multitemporal common structure to the abundance difference maps can remove the noise in some degree, highlight the structure of changes, and increase the contrast of changes and unchanged regions (as shown in strategy will achieve the enhancement of abundance the green box). These results illustrate that the coguidance difference maps and complete the full use of multitemporal spatial information. Fig. 4 gives the detection results obtained by different methods and the corresponding quantitative results are presented in Table I A combined analysis of Fig. 4 and Table I reveals that PCA–CVA loses some subpixel-level information due to its poor spectral utilization, resulting in a relatively low accuracy. The performance of BM is relatively poor when compared with other methods because its result depends on the unmixing model and three constraint conditions. These conditions may sensitive to some subtle changes. SU that obtains the change map by subtracting the corresponding abundance maps indicates the feasibility of unmixing method. Subtraction operation will perform better because the farmland data set has few spectral differences. The F-score (reaches on 95.04%) proves its effectiveness. JUWC achieves CD by a joint unmixing strategy, thresholding the abundance difference map. Due to its limited spatial utilization, many changes are incomplete and have some spectral singularities [as shown in the red box in Fig. 4(d)]. The proposed method (JUC) is the optimal result (F-score reached on 95.39%). This result indicates that the proposed coguidance strategy is an effective idea for full exploitation of spatial information. Meanwhile, it verifies that the injection of spatial information can improve the final results. Further comparing Fig. 4(d) and (e), the results obtained with no coguidance and with coguidance, respectively, it reveals that the major contribution of the coguidance approach is to further enhance the integrity of the changes and remove much of the spectral noise that is unavoidable in subpixel-level analysis.

B. Results and Discussion of River Data Set

In many practical situations, the acquisition of multitemporal images is often affected by some factors such as light conditions and seasons, which make the detection of actual changes very difficult. Additionally, the limited spatial resolution of MHSIs makes the visual interpretation is time consuming and subjective. Therefore, it is difficult to obtain a great result. To illustrate that the proposed method can solve the above problem, we test two hyperspectral data sets (river and lake data set) containing seasonal and light differences. The actual changes of river data set include river and some vegetations. The parameters of all methods are optimal as well. For the proposed method, the number of endmembers is three, meanwhile, $r = 3$, $\varepsilon_1 = 10^{-4}$, $\beta = 30$, $n = 20$, th $= 0.25$. The running time of four comparisons and JUC is 10, 85, 48, 20, and 30 s, respectively. The corresponding results are displayed in Fig. 5 and Table II.

The advantages of our proposed framework are not clear enough in the farm data set, where spectral variability is low. The complex background contained in the river data set may

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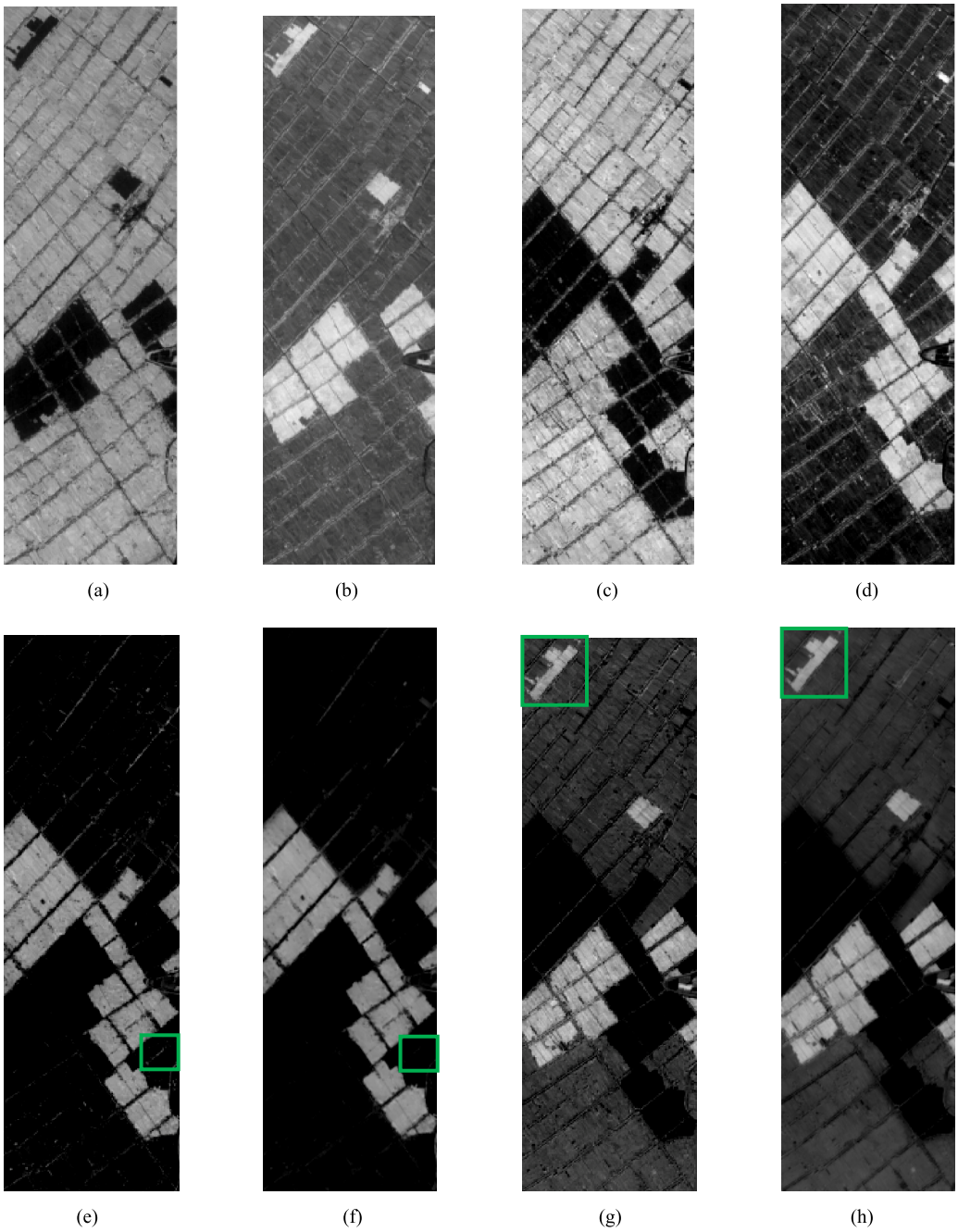


Fig. 3. Partial comparison results of joint unmixing and coguidance. (a) Abundance image of the first temporal image corresponding to the first endmember. (b) Abundance image of the first temporal image corresponding to the fourth endmember. (c) Abundance image of the second temporal image corresponding to the first endmember. (d) Abundance image of the second temporal image corresponding to the fourth endmember (e) Abundance difference image of different temporal image corresponding to the first endmember. (f) Result of (e) after coguidance. (g) Abundance difference image of different temporal image corresponding to the fourth endmember. (h) Result of (g) after co-guidance.

limit many methods. A comprehensive analysis of Fig. 5 and Table II shows that PCA–CVA has poor performance, when the data set contains the spectral variabilities [as shown in the green box in Fig. 5(a)], mainly due to its low utilization of spectral information. The result obtained by this method also further illustrates that the pixel-level method needs to be improved and extended to be more applicable MHSIs. The SU and BM method perform subpixel-level analysis based on the sparsity of MHSIs and three conditions, respectively, which improve the detection accuracy to some extent (F-score reaches on 78.85% and 72.26%), but they still suffer from the spectral variability. The accuracy of JUWC, modified within the existing framework, is high (F-score is 95.07%). Therefore, this result demonstrates the advantages of joint unmixing in MHSIs CD. Based on JUWC, we inject the common structure of multitemporal images, which can be found to further improve the detection accuracy (up to 95.96%). The experimental results of this data set also support the

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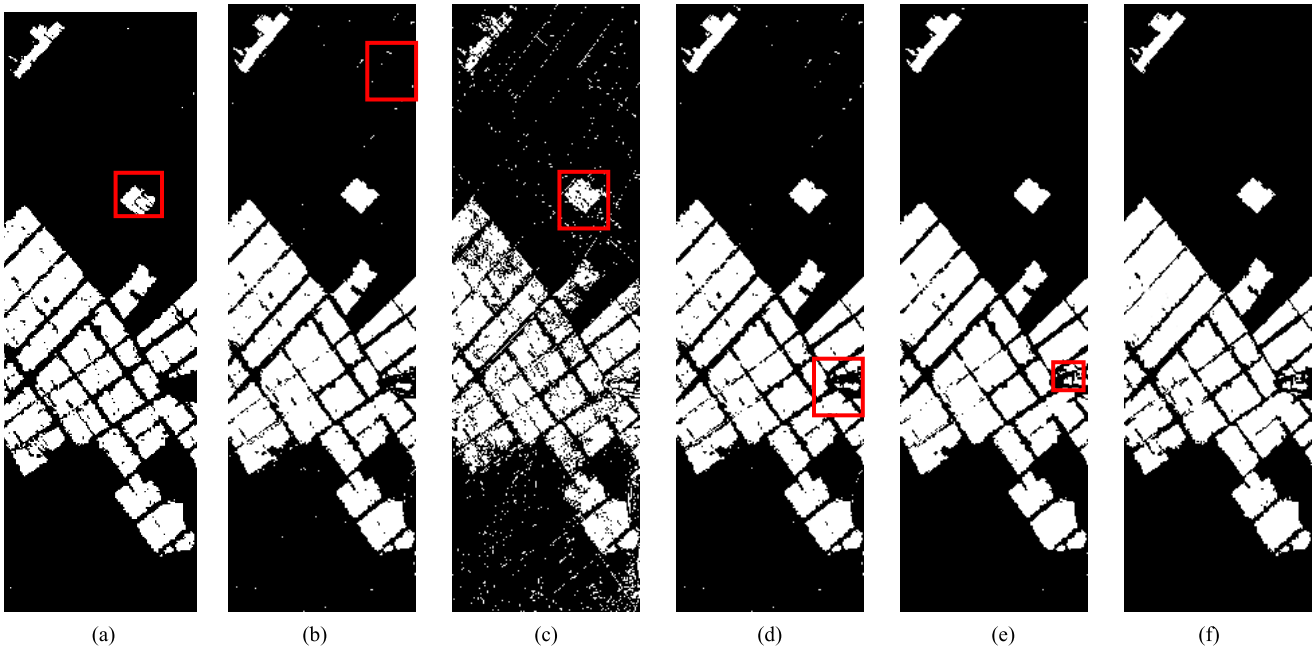


Fig. 4. Detection results of different methods on Farmland data set. (a) PCA-CVA. (b) SU. (c) BM (d) JUWC. (e) JUC. (f) Ground truth.

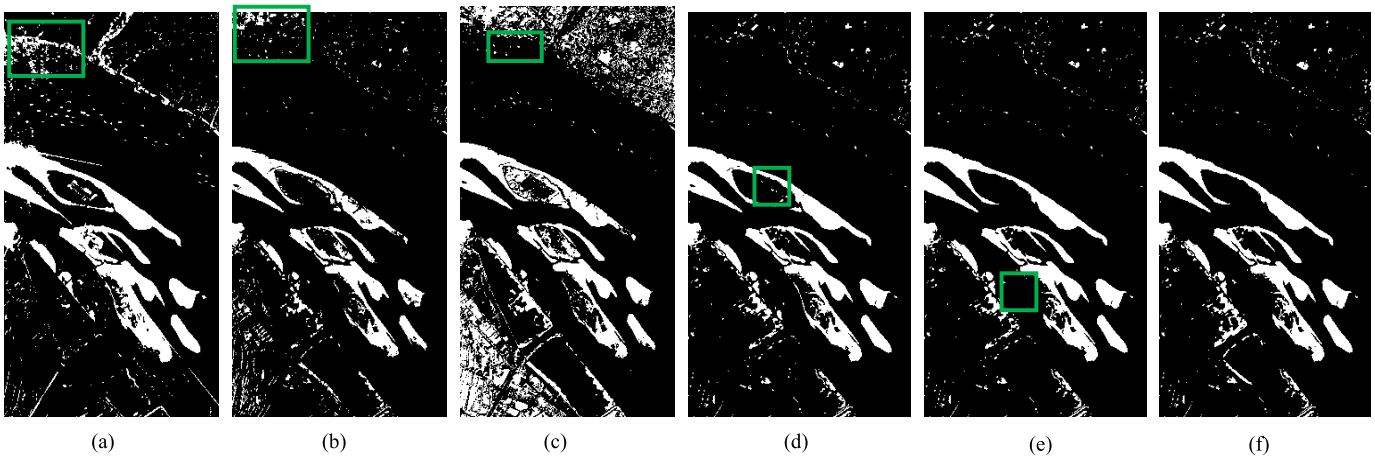


Fig. 5. Detection results of different methods on river data set. (a) PCA-CVA. (b) SU. (c) BM (d) JUWC. (e) JUC. (f) Ground truth.

TABLE II
QUANTITATIVE COMPARISON OF RIVER DATA SET CD RESULTS BY DIFFERENT METHODS

Method	Precision	Recall	F-score
PCA-CVA	62.42%	84.90%	71.94%
SU	81.34%	76.50%	78.85%
BM	56.45%	**98.78%**	72.26%
JUWC	96.17%	94.01%	95.07%
The proposed	**97.77%**	**94.21%**	**95.96%**

conclusions we mentioned, including JUC which is robust to spectral variability and enables to enhance the contrast between the changed and unchanged regions, and removal of noise.

C. Results and Discussion of Lake Data Set

To illustrate the suitability of our proposed method further, we introduce a lake data set, which still includes the effects of spectral variability and season. The actual changes of lake data set contain lake, river, and crops. The parameters of all comparison methods are optimal. For the proposed framework, the number of endmembers is three, and the relevant parameters of the coguidance scheme are set to: $r = 3$, $\varepsilon_1 = 10^{-4}$, $\beta = 30$, $t = 30$, th $= 0.15$. The running time of PCA-CVA, SU, BM, JUWC, and JUC is 5, 55, 23, 8, and 18 s, respectively. The corresponding processing results and detection results are displayed in Figs. 6 and 7 and Table III.

Fig. 6 shows comparisons of abundance results corresponding to each endmember of MHSIs. Meanwhile, the abundance difference images before and after the injection of multitemporal spatial information are displayed in Fig. 6 as well. The following two conclusions can be found. On the one hand, for data sets with severe spectral variability, the strategy of joint unmixing is very effective. There is an evident that the pixel value differences of corresponding abundance maps are relatively small (marked by red

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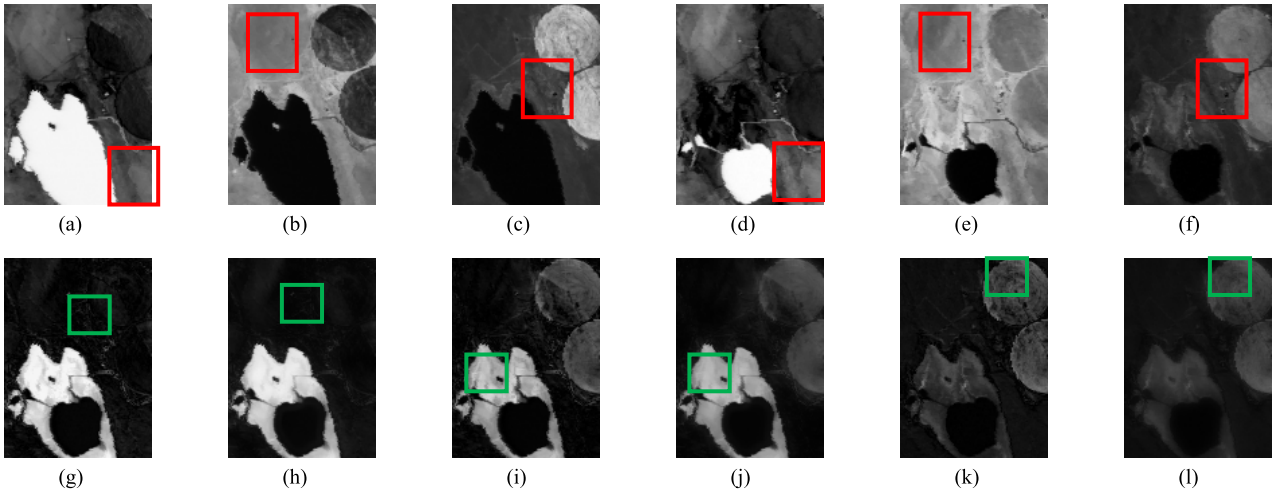


Fig. 6. Comparison results of joint unmixing and coguidance unmixing. (a)–(c) Abundance image of the first temporal image corresponding to different endmembers. (d)–(f) Abundance image of the second temporal image corresponding to different endmembers. (g) Abundance difference image of different temporal image corresponding to the first endmember. (h) Result of (g) after coguidance. (i) Abundance difference image of different temporal image corresponding to the second endmember. (j) Result of (i) after coguidance. (k) Abundance difference image of different temporal image corresponding to the third endmember. (l) Result of (k) after coguidance.

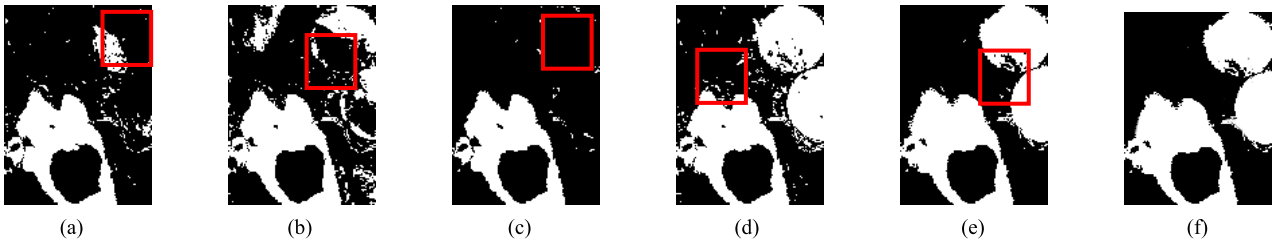


Fig. 7. Detection results of different methods on Lake data set. (a) PCA-CVA. (b) SU. (c) BM (d) JUWC. (e) JUC. (f) Ground truth.

TABLE III
QUANTITATIVE COMPARISON OF LAKE DATA SET CD RESULTS BY DIFFERENT METHODS

Method	Precision	Recall	F-score
PCA-CVA	94.33%	54.45%	69.04%
SU	85.56%	70.40%	77.24%
BM	**99.17%**	47.31%	64.06%
JUWC	94.45%	94.81%	94.63%
The proposed	**98.72%**	**98.15%**	**98.43%**

boxes in Fig. 6). Therefore, the spectrum of the suspected changed regions is easy to be highlighted in the abundance difference map [Fig. 6(g), (i), and (k)]. On the other hand, the injection of multitemporal common structure can further highlight the changes structure [as shown in the green boxes in Fig. 6]. It is evident that the coguidance strategy is equally effective.

The detection results obtained by different methods are given in Fig. 7 and the results of quantitative analysis are given in Table III. In general, we consider all three values of the quantification to be high, indicating that the results are optimal. It can be found that BM method has relatively poor performance, although its precision is high, but its recall is relatively low, only 47.31%. The main reason is that the method is first unmixing separately, and then, introduces the thresholding conditions to achieve CD. When the spectral variability is severe, it is difficult to obtain good results from this unmixing way, regardless of the subsequent threshold conditions, unmixing errors have been accumulated. PCA–CVA method is a pixel-level method in which the rich spectral information is lost during processing. Hence, the subtle changes in the spectrum cannot be detected, resulting in the detection accuracy is not satisfactory. SU adopts a multitemporal hyperspectral image stacking way and extracts the abundance difference maps based on the spectral libraries, which increases the utilization of the spectra to a certain extent and avoids the error caused by the separate unmixing, but the low utilization of the multitemporal spatial information make the F-score is only 77.24%. By visual interpretation of Fig 7, it can be found that the adopted joint unmixing method can eliminate the spurious variation caused by spectral variability and improve the detection accuracy (F-score has reached on 94.63%), but there are still some discrete false alarm areas [as shown in the red box of Fig. 7 (d)]. The proposed method achieves the optimal results (F-score reaches on 98.43%), with improved accuracy compared to JUWC. The main reason for obtaining this result is that if there is a constraint of the multitemporal spatial information. The multitemporal spatial information is hardly considered in JUWC. On the contrary, JUC can achieve the injection of spatial information to improve the accuracy.

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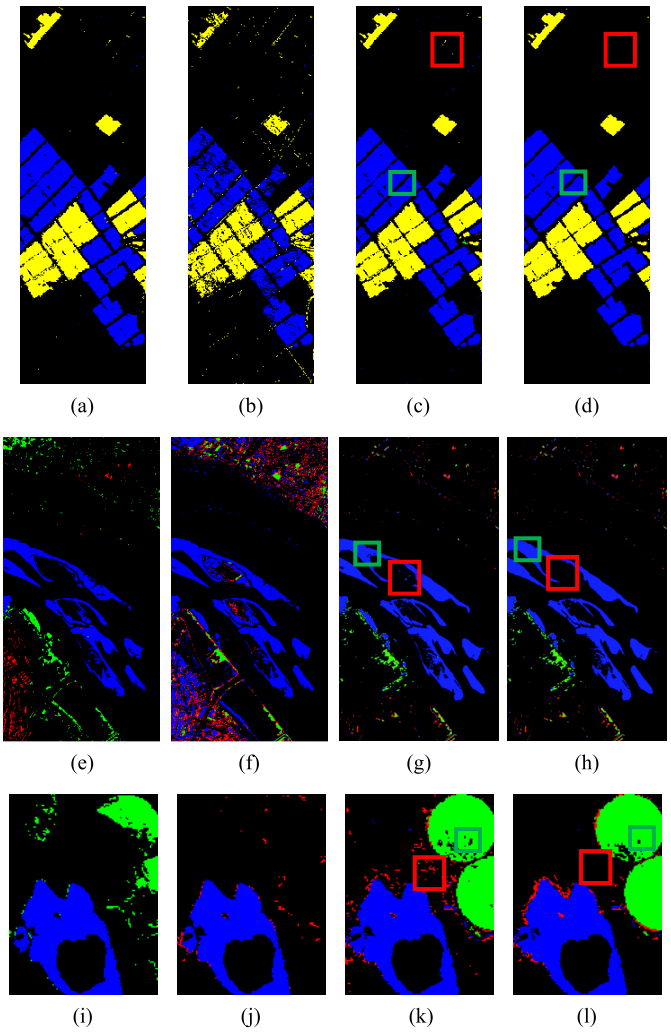


Fig. 8. Multiclass changes results of different methods on the adopted data set. (a) Result of farmland data set by SU. (b) Result of farmland data set by BM. (c) Result of farmland data set by JUWC. (d) Result of farmland data set by JUC. (e) Result of river data set by SU. (f) Result of river data set by BM. (g) Result of river data set by JUWC. (h) Result of river data set by JUC. (i) Result of lake data set by SU. (j) Result of lake data set by BM. (k) Result of lake data set by JUWC. (l) Result of lake data set by JUC.

D. Discussion of Multiclass Changes

Binary CD is mainly to detect the changing locations. In many cases, however, obtaining the change classes can further enrich the CD results. The higher spectral resolution of MHSIs makes it possible to make full use of spectral information to detect multiclass changes. According to the principle of spectral unmixing, multiclass CD can be achieved in a completely unsupervised manner. The detection results are displayed in Fig. 8.

The results of multiclass changes obtained based on the different unmixing schemes are shown in Fig. 8. According to the analysis, considering different endmember corresponding to different class, it is possible to obtain multiclass changes by assigning different labels to the positions corresponding to different class. PCA–CVA that mainly adopts the spatial and temporal information rather than unmixing strategy; therefore, we do not provide the multiclass changes based on this method. Both the SU and BM methods suffer from bias in class judgments due to the effects of abundance difference maps.

TABLE IV
NOTATIONS USED IN THIS PAPER

Symbol	Descriptions
K	The number of endmembers
T	The number of images contained in multi-temporal sequence
N	The number of pixels in each temporal image
x_{it}	The ith pixel in the tth image
a_{kit}	Abundance of the kth endmember in the ith pixel of the tth image
e_{kt}	The kth endmember of tth image
de_{kt}	The spectral perturbation of tth temporal image k^{th} endmember
b_{it}	The noise of the ith pixel in the tth image
X_t	Matrix form of all temporal images
E_t	Endmembers matrix
dE_t	The spectral perturbation matrix
B_t	Noise matrix
σ	Energy value of the spectral variability
κ	Instantaneous energy variation of the spectral variability
M	Endmember sets
$\wp_A$	Abundance sets
P_t	Spectral perturbation sets
Ψ	The regularization term
τ	The regularization parameter
F	Normalized difference measure function
H	The loss function
R	The least-squares regression function
I	Abundance difference map
G	First principal component map
a	Weight and bias coefficient
$N(p)$	The block centered at the point p
R	The number of blocks
$\lvert N\rvert$	Total pixel number in block $N(p)$
$e\left(I_p, G_p\right)^2$	Similarity measure function within a block
G	The sum function all blocks
$C(I)$	Solving constraint functions
I_p	The pixel value of position p in the enhanced abundance maps
$I_{0,p}$	The pixel value of position p in I
β	The constraint parameter
S	Smoothing term
ε_1	Smoothing parameter
F	Co-guidance objective function
$I_{[i,k]}$	The enhanced abundance map after processing by i^{th} image
$I_{s,k}$	The enhanced abundance map of kth endmember
N	Iteration number of solving f
$\mu(\bullet)$	The pixel mean value in the block $N(p)$
$M_I^{(n)}$	The first coefficient for the nth iteration
$J_I^{(n)}$	The second coefficient for the nth iteration
$K_I^{(n)}$	The third coefficient for the nth iteration
B	The parameter of $J_I^{(n)}$ and $K_I^{(n)}$
B_{CM}	Binary change result
$th()$	Thresholding function
M_{CM}	Multiple change result

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By analyzing the accuracy of each method in the above parts, the proposed method has the best performance [Fig. 8(d), (h), and (l)]. Further analysis of JUWC and JUC reveals that the guided injection of the multitemporal common structure can remove many of the spurious changes (as shown in the red box of Fig. 8) and the structure of the changes is more integrated (as shown in the green box of Fig. 8). Additionally, it can be observed that the farmland data set consists mainly of three types of crop changes; river data set includes mainly three types of changes; lake data set contains mainly three types of changes. Actually, the correctness of the types of changes determined by the results of the individual unmixing and CD method. Thus, the proposed joint unmixing and coguidance scheme is very necessary.

IV. Conclusion

CD for hyperspectral image is one of the most important topics for multitemporal image analysis. It is aim to identify the strong and subtle changes both from complex background. In this article, a novel unsupervised framework based on multitemporal joint unmixing and multitemporal spatial information coguidance (JUC) for MHSIs CD is proposed to achieve temporal, spatial, and spectral feature joint analysis. It adopts a joint unmixing method to address the influence on the severe spectral variabilities and transfers to the field of CD. Further, a new multitemporal spatial information coguidance idea is proposed to combine the spatial information of each temporal principal component into the abundance difference maps obtained by joint unmixing. This processing way not only highlights the structure of changes in abundance difference maps, but also remove the noise. Three data sets, including multitemporal farmland data set, river data set, and lake data set, are used to verify the effectiveness and superiority of the proposed method. Meanwhile, in contrast to the other popular CD methods, the experimental results show that our proposed method has the more robust and stable ability to CD. This can be owing to that joint unmixing eliminate the effects of spectral variability, meanwhile, the coguidance increases the utilization of multitemporal spatial information.

Appendix

The notations used in this article are listed in Table IV.

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